
GENERAL RELATIVITY

Tutorial problem set 15, 10.01.2020.

■ **PROBLEM 37** Dimming of objects falling into a black hole.

Consider an observer outside of a Schwarzschild black hole sitting at constant Schwarzschild coordinates $(r_0, \vartheta = \frac{\pi}{2}, \varphi = 0)$. At some point $t = 0$ he releases a beacon that starts falling towards the black hole, while constantly emitting a monochromatic light signal (of frequency ν_{em}) back to the observer (in the radial direction). In this problem you will determine what is the signal received by the observer.

- (a) One generally needs to be careful what kind of velocities one defines in GR. Here we can define at least three ones for the beacon of certain use:
- (i) The coordinate velocity, $v = \frac{dr}{dt}$, that measures what is the coordinate distance dr that the beacon travels during coordinate time dt ,
 - (ii) The r-component of the 4-velocity, $u_r = \frac{dr}{d\tau}$, that measures the coordinate distance dr the beacon travels during its own proper time interval $d\tau$,
 - (iii) The 'velocity of a stationary observer', $w_R = \frac{dr}{d\tau_R}$, which is the velocity of the beacon that would be measured by an observer at the fixed coordinate R , at the moment the beacon is passing him by, where τ_R is the proper time of that observer.

Calculate these three velocities as functions of r . Rather than solving the geodesic equations, make use of the Schwarzschild line element, and the conserved quantity E derived in problem 32 of the tutorial problem set 12. Fix E by imposing initial conditions.

- (b) Find the relation between the coordinate time and the proper time of a static observer at R , and between coordinate time and the proper time of the beacon at r .

In order to find the frequency of the signal received by the observer at r_0 , we will employ a similar strategy as in problem 34 of the tutorial problem set 13. We will relate the proper time (of the beacon) between the emission of the two successive light crests, and the proper time of the observer at r_0 between the absorption of these two crests.

- (c) The beacon emits the first crest at (t_{em1}, r_{em1}) , and the second one (t_{em2}, r_{em2}) . The coordinate time between the emission of two successive crests is $\Delta t_{em} = t_{em2} - t_{em1}$, and during that time the beacon has moved by $\Delta r_{em} = r_{em2} - r_{em1}$. Calculate the coordinate time Δt_{re} between the reception of the two crests by the observer at r_0 . In order to do that, calculate the coordinate time it takes each crest to reach the observer.
- (d) We can safely assume that the time interval between the emission of the two crests is much smaller than any other relevant scale. Expand the expression obtained in (c) to linear order in Δt_{em} , and express the ratio $\Delta t_{re}/\Delta t_{em}$ in terms of r_0 , $R_S = 2MG_N$ and $r_{em1} \equiv r$.

- (e) Using the expressions found in (b) and (d) show that the frequency received by the observer is

$$\nu_{re} = \nu_L \sqrt{\frac{A(r)}{A(r_0)}} \times \frac{\sqrt{A(r)}}{\sqrt{A(r_0)} + \sqrt{A(r_0) - A(r)}} , \quad (37.1)$$

where $A(r) = 1 - \frac{R_s}{r}$. Compare this result to the one obtained in problem 34 of the tutorial problem set 13. Interpret where does the extra term here come from.

■ **PROBLEM 38** Motion of a particle in Kerr space-time.

The line element of the Kerr space-time in Boyer-Lindquist coordinates is

$$ds^2 = - \left[1 - \frac{2MG_N r}{\rho^2} \right] dt^2 - \frac{4MG_N a r \sin^2 \theta}{\rho^2} dt d\phi + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 \\ + \left[r^2 + a^2 + \frac{2MG_N r a^2 \sin^2 \theta}{\rho^2} \right] \sin^2 \theta d\phi^2 , \quad (38.1)$$

where

$$\rho^2 = r^2 + a^2 \cos^2 \theta , \quad \Delta = r^2 - 2MG_N r + a^2 . \quad (38.2)$$

In this problem you will examine the orbits of massive particles and light particles that lie in the equatorial plane $\theta = \frac{\pi}{2}$.

- (a) Write the metric (38.1) in the equatorial plane $\theta = \frac{\pi}{2}$.
(b) Since the metric (38.1) does not depend on coordinates t and ϕ we can immediately conclude that there are two Killing vectors in this space-time,

$$K^\mu = (1, 0, 0, 0) , \quad R^\mu = (0, 0, 0, 1) , \quad (38.3)$$

associated to time translation symmetry and axial symmetry, respectively. When there are Killing vectors, there are conserved quantities along geodesics associated to them. In this case those are energy and the z -component of the angular momentum,

$$E = -K_\mu \frac{dx^\mu}{d\lambda} , \quad \ell = R_\mu \frac{dx^\mu}{d\lambda} . \quad (38.4)$$

Express $dt/d\lambda$ and $d\phi/d\lambda$ in terms of E and ℓ and the radial coordinate.

- (c) Using results from (a) and (b) find the effective potential for the motion of a massless particle in the equatorial plane,

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + W_{\text{eff}}(E, \ell; r) = \frac{E^2}{2} . \quad (38.5)$$

- (d) Find the effective potential for massive particles (take $\lambda = \tau$),

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 + V_{\text{eff}}(E, \ell; r) = \frac{E^2 - 1}{2} . \quad (38.6)$$

- (e) Consider a massive particle initially infalling radially ($\ell = 0$) towards the Kerr black hole, and with negligible initial kinetic energy ($E = 1$ – why this value?). Kerr space-time is stationary, but it is not static (black hole is rotating), and as an effect it will drag the initially radially infalling particle with its rotation, i.e. $d\phi/dr$ will not be constant. Show that

$$\frac{d\phi}{dr} = -\frac{2MG_N a}{r\Delta} \left[\frac{2MG_N}{r} \left(1 + \frac{a^2}{r^2} \right) \right]^{-1/2}. \quad (38.7)$$

The angle $\Delta\phi(r_0)$ the trajectory of the particle has swept from $r = \infty$ to some $r = r_0$ can in principle be calculated by integrating this expression from ∞ to r_0 . Will this particle eventually fall into the black hole?

- (f) In the limit of extremal Kerr black hole where $a = MG_N$, find the two possible circular light-like orbits. Are they stable or unstable? What are their angular velocities?

There are two circular photon orbits in the non-extremal case as well,

$$r_1 = 2MG_N \left\{ 1 + \cos \left[\frac{2}{3} \arccos \left(-\frac{|a|}{MG_N} \right) \right] \right\}, \quad (38.8)$$

$$r_2 = 2MG_N \left\{ 1 + \cos \left[\frac{2}{3} \arccos \left(\frac{|a|}{MG_N} \right) \right] \right\}. \quad (38.9)$$

Convince yourself that they reduce to the right limit when $a \rightarrow 0$.